

Damla Altunkaya

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Education

Bogazici University , Bachelor of Science (B.Sc.) in Physics	Sept 2021 – June 2026
• GPA: 3.15/4.0	
Bornova Anatolian High School , Math-Science	Sept 2017 – June 2021
• GPA: 94.17/100	

Research Interests

- Quantum Thermodynamics: *Entanglement Entropy*
- Condensed Matter Physics: *Hubbard Model, Tight Binding Model, Heat Equation, Discrete Schrodinger Equation*
- High Energy Physics: *Supersymmetric Theories, Localization in Path Integrals, Scattering Amplitudes*

Research Experience

The Effect of External Magnetic Field on Entanglement Entropy	Oct 2020 - May 2021
Supervisors: Omer C. Gurdogan - University of Southampton, Ceren B. Dag - Harvard University	
• Held weekly meetings on topics ranging from the very fundamentals of quantum mechanics to quantum thermodynamics. • Chose the XXZ Heisenberg model to incorporate the effects of an external magnetic field within the Hamiltonian. • Computed density matrix and the corresponding entropy for each value of the external magnetic field.	
The Restricted Three Body Problem	July 2024 - Sept 2024
Supervisor: Kursat Yilmaz - University of Toledo	
• Introduced rotating frame and formulated the corresponding Lagrangian of the system. • Derived equations of motion and effective potential; computed equilibrium points (Euler-Lagrange points). • Discussed allowed and forbidden regions, and Lyapunov orbit in detail.	
The Hubbard Model	Feb 2025 - June 2025
Supervisor: Levent Akant - Bogazici University	
• Made weekly presentations starting from creation-annihilation operators to AFM-FM properties. • Plotted DoS and Green's functions of various lattice models to visualize theoretical results such as Mott gap and support discussion. • Investigated each component of the Hubbard model Hamiltonian in both site space and momentum space to acquire intuition and more significantly exactly diagonalize the Hamiltonian. • Computed magnetic moment in strong coupling limit, obtained ground state energy and exchange coupling J to discuss the Ferromagnetic and Antiferromagnetic properties of matters.	
Number Theoretical Approach to Scattering Amplitudes	July 2025 - Aug 2025
Supervisor: Omer C. Gurdogan - University of Southampton	
• Demonstrated that Feynman integrals yield polylogarithms and Multiple Zeta Values (MZVs) as their transcendental results in multiloop amplitudes. • Performed Hopf algebra operations on MZVs including symbol calculus, shuffle/stuffle products, and the coaction mechanism utilizing the HyperLogProcedures package in Maple and Mathematica. • Applied a map operation to extract single-valued (SV) versions of the MZVs and subsequently obtained dimensions of particular weights. • Demonstrated the antipodal duality between the A-form and F-form representations to reveal the underlying algebraic symmetries of the scattering amplitude symbol.	

Localization in Path Integrals

Sept 2025 - Jan 2026

Supervisor: Can Kozcaz - Bogazici University

- Studied the fundamentals of supersymmetric field theories and Supersymmetric Quantum Mechanics (SQM), primarily following David Tong's lecture notes.
- Explored the mechanism of topological twisting in $\mathcal{N} = (2, 2)$ theories to transform physical Lagrangians into topological ones, utilizing Hori's Mirror Symmetry.
- Investigated the localization principle, focusing on how the saddle-point approximation yields exact results for certain path integrals as introduced in Stefano Cremonesi's work.

Solutions to Heat Equation and Discrete Schrödinger Equation

Oct 2026 - Ongoing

Supervisor: Levent Akant- Bogazici University

- Investigated the analytical solutions of the discrete Schrödinger equation and the heat equation on a 1D infinite lattice, utilizing the properties of modified Bessel functions.
- Calculated the thermodynamic potential for the Tight Binding model by applying Mellin transform techniques.
- Developed high-temperature expansions of the partition function through the heat kernel expansion.

Scholarships

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| • National Undergraduate Scholarship (2205) by TUBITAK | 2021-2026 |
| • Outstanding Success Scholarship by Turkish Educational Foundation (TEV) | 2021-2026 |

Oral Presentations

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| • <i>The Restricted Three Body Problem</i> , DRP Turkey Symposium, Sept 6th 2024, Sabanci University | youtube.com/3body-problem |
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Conferences and Schools

- **Many-Particle Systems: From Condensed Matter to Quarks and Stars**, Bogoliubov Laboratory of Theoretical Physics-JINR, Dubna, Feb 2024
- **Fields and Strings**, Steklov Mathematical Institute, Moscow, Feb 2024
- **From Light to Dark**, Bilkent University, Ankara, July 2025
- **Mathematical Physics Summer School**, SwissMAP, Les Diablerets, Aug 2025

Accepted & Not Attended *due to personal/scheduling reasons*

- JINR START Programme VBLHEP (funded)
- EPFL Workshop on Duality, Geometry, Physics (funded)

Independent Coursework

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| • Quantum vs. Thermodynamics, Onur Pusuluk, Sciences Village, Izmir/Turkey | Aug 2024 |
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Selected Coursework

- **PHYS312 - Modern Physics:** Foundations of QFT, Quantum Electrodynamics (QED), gauge fixing and Faddeev-Popov ghosts, perturbation theory, spontaneous symmetry breaking
- **PHYS48A - Path Integral and Its Applications:** Feynman path integrals, propagators (free particle, harmonic oscillator), Born approximation, Wick's theorem (Gaussian integrals, contractions/pairings), WKB approximation, semiclassical (saddle-point) methods, instantons and tunneling
- **PHYS585 - Introduction to Quantum Field Theory:** QFT foundations (Klein-Gordon equation, real and complex scalar field quantization), Dyson/Feynman-Dyson expansion and time-ordering, Feynman diagrams and rules, LSZ reduction, scattering amplitudes and cross sections, scalar QED (gauge interaction basics)

Teaching Experience

Student Teaching Assistance, Bogazici University

Feb 2024 – May 2024

- Equilibrium, Gravitation, Fluid Mechanics, Wave Mechanics and Thermodynamics Lab Assistant

Student Teaching Assistance, Bogazici University

Oct 2024 – Jan 2025

- Electricity and Magnetism Lab Assistant

Computer Skills

Advanced knowledge: Python, ROOT, $\text{\LaTeX}$

Basic knowledge: Mathematica, Maple

Language Skills

Turkish: Native

English: Fluent